

Electrical Instrumentation Part-II

PowerPack Phase 3

Active Filters, State Variable Filters and Switched-Capacitor Filters

Reverse-engineered semester preparation notes

Diagram-safe, derivation-first, exam-answer style

Semester Command

This Phase 3 PowerPack is for mastering filter questions. Learn every answer in this order:

Definition → Standard transfer function → Diagram → Parameter comparison → Derivation or de

In every diagram, use straight and perpendicular lines. Keep labels far away from components. If a label becomes long, split it into two or three lines inside a box.

Contents

1	Phase 3 Target: What You Must Be Able To Answer	4
2	Grassroot Filter Concept	5
2.1	Four Basic Filter Types	5
2.2	Basic Frequency Response Diagram	6
3	Standard Second-Order Filter Forms	7
3.1	Why Second-Order Form is Important	7
3.2	Standard Transfer Functions	7
4	Butterworth Filter	9
4.1	Definition	9
4.2	Properties of Butterworth Filter	9
5	Proof: Butterworth Poles Lie on Unit Circle	9
5.1	Starting Point	9
5.2	Butterworth Pole Diagram	11
6	Chebyshev Filter	12

6.1	Definition	12
6.2	Properties of Chebyshev Filter	12
7	Proof: Chebyshev Poles Lie on Ellipse	12
7.1	Chebyshev Pole Diagram	14
8	Why Normalized Chebyshev -3 dB Frequency is Greater than 1 rad/sec	15
8.1	Derivation of Exact -3 dB Frequency	16
9	Butterworth vs Chebyshev: Semester Comparison	17
10	Second-Order Band-Pass Filter Design	18
10.1	Standard Form	18
10.2	Numerical Exam Template	18
10.3	Band-Pass Response Diagram	19
11	Band-Stop Filter from Band-Pass Filter	20
11.1	Band-Stop Realization Diagram	20
12	State Variable Filter	21
12.1	Definition	21
12.2	Why It is Called Universal Filter	21
12.3	State Variable Block Diagram	21
13	State Variable Transfer Functions	22
13.1	Comparison with Standard Denominator	22
14	Switched-Capacitor Circuits	24
14.1	Definition	24
14.2	Derivation of Equivalent Resistance	24
14.3	Resistor to Switched-Capacitor Diagram	25
15	Advantages of Switched-Capacitor Circuits	26
16	Active RC to Switched-Capacitor Conversion	27
16.1	Basic Rule	27
16.2	Example	27
16.3	Clean Switched-Capacitor Active Filter Idea	28

17 Low-Pass Filter Design by Butterworth Approximation	29
17.1 Order Calculation	29
17.2 Second-Order Butterworth Transfer Function	29
17.3 Sallen-Key Low-Pass Conceptual Diagram	30
18 High-Pass Filter Concept and Switched-Capacitor Version	31
18.1 Second-Order High-Pass Standard Form	31
18.2 Active RC High-Pass Conceptual Diagram	31
19 Exam Design Routine	32
19.1 Routine for Band-Pass Numerical	32
19.2 Routine for Band-Stop Numerical	32
19.3 Routine for Chebyshev Order	32
19.4 Routine for Switched-Capacitor Conversion	32
20 One-Page Formula Bank	34
21 How to Recognise Filter Questions in Semester	35
22 Final Rapid Revision for Phase 3	36
22.1 If Only 30 Minutes Are Left	36
22.2 Most Probable Diagram Commands	36

1. Phase 3 Target: What You Must Be Able To Answer

After finishing this phase, you must be able to answer these semester questions:

1. Explain active filters and write standard second-order transfer functions.
2. Prove that Butterworth poles lie on a unit circle.
3. Prove that Chebyshev poles lie on an ellipse.
4. Explain why normalized Chebyshev -3 dB cut-off is greater than 1 rad/sec .
5. Design second-order low-pass, high-pass, band-pass and band-stop filters.
6. Explain state variable filter and why it is called universal filter.
7. Realize band-pass and band-stop filters using state variable representation.
8. Explain switched-capacitor circuits and derive $R_{\text{eq}} = \frac{1}{f_s C_s}$.
9. Convert active RC filters into switched-capacitor filters.
10. Solve numerical design questions using ω_0 , B , Q , gain and cut-off frequency.

Must Know

Filter questions are formula plus diagram based. If you know the standard transfer function and can compare coefficients, you can solve most numerical questions.

2. Grassroot Filter Concept

A filter is a frequency-selective circuit. It allows some frequency components to pass and attenuates other frequency components.

Filter = frequency selector

An **active filter** uses active devices such as op-amps along with resistors and capacitors.

Active filter = op-amp + R + C

Active filters are preferred because they can provide gain, avoid inductors and are suitable for integrated-circuit realization.

2.1. Four Basic Filter Types

Filter Type	Meaning
Low-pass filter	Passes low frequencies and attenuates high frequencies.
High-pass filter	Passes high frequencies and attenuates low frequencies.
Band-pass filter	Passes a band of frequencies around center frequency ω_0 .
Band-stop filter	Rejects a band of frequencies around center frequency ω_0 .

2.2. Basic Frequency Response Diagram

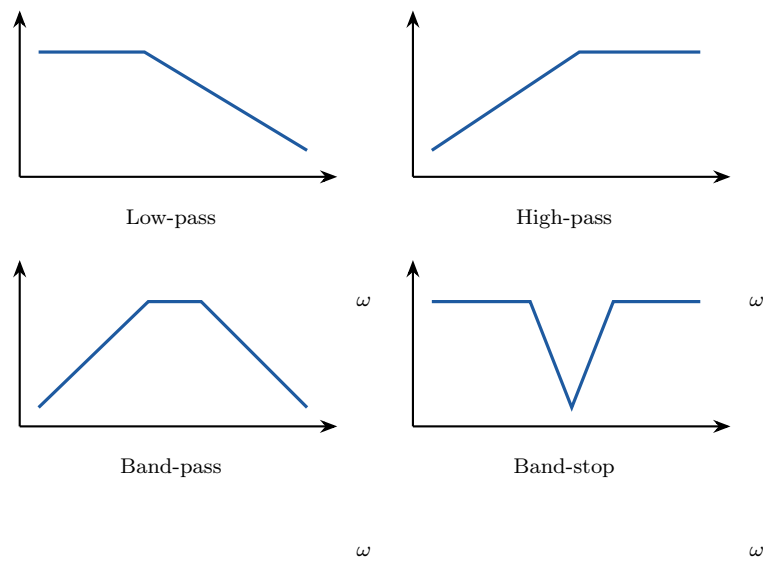


Figure 1: Basic response shapes of four filter types.

3. Standard Second-Order Filter Forms

3.1. Why Second-Order Form is Important

Most semester filter questions reduce to comparing a given transfer function with a standard second-order form.

The standard denominator is

$$s^2 + \frac{\omega_0}{Q}s + \omega_0^2$$

where,

ω_0 = center or natural angular frequency

Q = quality factor

and

B = bandwidth

For band-pass and band-stop filters,

$$B = \frac{\omega_0}{Q}$$

Therefore,

$$Q = \frac{\omega_0}{B}$$

3.2. Standard Transfer Functions

$$H_{LP}(s) = \frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$H_{HP}(s) = \frac{Ks^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$H_{BP}(s) = \frac{KBs}{s^2 + Bs + \omega_0^2}$$

$$H_{BS}(s) = \frac{K(s^2 + \omega_0^2)}{s^2 + Bs + \omega_0^2}$$

Must Know

For band-pass and band-stop numerical questions, use:

$$B = \frac{\omega_0}{Q}$$

and

$$s^2 + Bs + \omega_0^2$$

as the denominator.

4. Butterworth Filter

4.1. Definition

A Butterworth filter is a filter whose magnitude response is maximally flat in the pass-band. It has no ripple in pass-band and no ripple in stop-band.

For normalized low-pass Butterworth filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

where,

n = order of filter

4.2. Properties of Butterworth Filter

1. Maximally flat pass-band.
2. No ripple in pass-band.
3. No ripple in stop-band.
4. Smooth monotonic response.
5. Poles lie on a circle in the s -plane.

5. Proof: Butterworth Poles Lie on Unit Circle

5.1. Starting Point

For normalized Butterworth low-pass filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

For pole calculation, replace $j\omega$ by s . The denominator equation becomes

$$1 + (-1)^n s^{2n} = 0$$

This gives $2n$ roots symmetrically located in the s -plane. The stable filter uses only left-half-plane poles.

The stable pole form is

$$s_k = -\sin \theta_k + j \cos \theta_k$$

where,

$$\theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, 2, \dots, n$$

Let,

$$s_k = \sigma_k + j\omega_k$$

Then,

$$\sigma_k = -\sin \theta_k$$

and

$$\omega_k = \cos \theta_k$$

Now,

$$|s_k|^2 = \sigma_k^2 + \omega_k^2$$

$$|s_k|^2 = (-\sin \theta_k)^2 + (\cos \theta_k)^2$$

$$|s_k|^2 = \sin^2 \theta_k + \cos^2 \theta_k$$

$$|s_k|^2 = 1$$

Therefore,

$$\boxed{|s_k| = 1}$$

Hence, normalized Butterworth poles lie on a unit circle in the s -plane.

5.2. Butterworth Pole Diagram

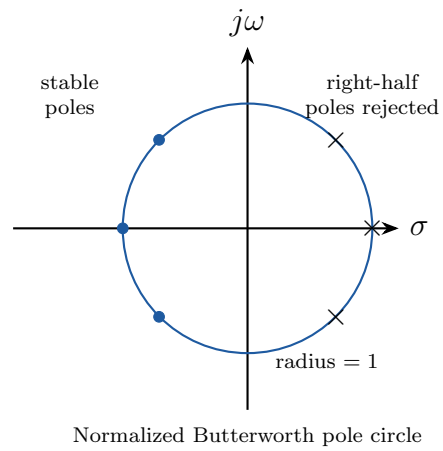


Figure 2: Butterworth poles lie on a unit circle; only left-half-plane poles are used.

6. Chebyshev Filter

6.1. Definition

A Chebyshev filter is a filter having ripple in the pass-band and sharper transition than Butterworth filter for the same order.

For normalized Chebyshev low-pass filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

where,

ϵ = ripple factor

and

$T_n(\omega)$ = Chebyshev polynomial of order n

6.2. Properties of Chebyshev Filter

1. Equiripple response in pass-band.
2. Sharper cut-off than Butterworth filter.
3. More selective for same order.
4. Poles lie on an ellipse in the s -plane.
5. The normalized frequency $\omega = 1$ is the end of ripple band, not always the -3 dB frequency.

7. Proof: Chebyshev Poles Lie on Ellipse

The stable Chebyshev poles can be written as

$$s_k = -\sinh \alpha \sin \theta_k + j \cosh \alpha \cos \theta_k$$

where,

$$\theta_k = \frac{(2k-1)\pi}{2n}$$

and

$$\alpha = \frac{1}{n} \sinh^{-1} \left(\frac{1}{\epsilon} \right)$$

Let,

$$s_k = \sigma_k + j\omega_k$$

Then,

$$\sigma_k = -\sinh \alpha \sin \theta_k$$

and

$$\omega_k = \cosh \alpha \cos \theta_k$$

Therefore,

$$\frac{\sigma_k}{\sinh \alpha} = -\sin \theta_k$$

and

$$\frac{\omega_k}{\cosh \alpha} = \cos \theta_k$$

Squaring and adding,

$$\left(\frac{\sigma_k}{\sinh \alpha} \right)^2 + \left(\frac{\omega_k}{\cosh \alpha} \right)^2 = \sin^2 \theta_k + \cos^2 \theta_k$$

$$\boxed{\left(\frac{\sigma}{\sinh \alpha} \right)^2 + \left(\frac{\omega}{\cosh \alpha} \right)^2 = 1}$$

This is the equation of an ellipse. Hence, Chebyshev poles lie on an ellipse.

7.1. Chebyshev Pole Diagram

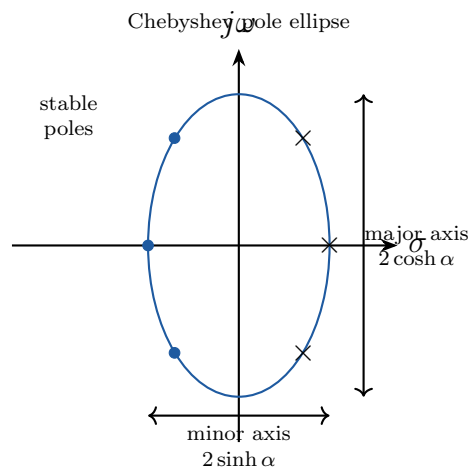


Figure 3: Chebyshev poles lie on an ellipse in the s -plane.

8. Why Normalized Chebyshev -3 dB Frequency is Greater than 1 rad/sec

For Chebyshev filter,

$$|H(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

At the end of the pass-band ripple region,

$$\omega = 1$$

and,

$$T_n(1) = 1$$

Therefore,

$$|H(j1)|^2 = \frac{1}{1 + \epsilon^2}$$

For usual Chebyshev ripple less than 3 dB,

$$\epsilon < 1$$

So,

$$1 + \epsilon^2 < 2$$

Therefore,

$$\frac{1}{1 + \epsilon^2} > \frac{1}{2}$$

But the -3 dB frequency is defined by

$$|H(j\omega_{3dB})|^2 = \frac{1}{2}$$

At $\omega = 1$, the magnitude has not yet fallen to $\frac{1}{2}$. Therefore, the -3 dB frequency must occur after $\omega = 1$.

$\omega_{3dB} > 1 \text{ rad/sec}$

8.1. Derivation of Exact -3 dB Frequency

At -3 dB,

$$\frac{1}{1 + \epsilon^2 T_n^2(\omega_{3dB})} = \frac{1}{2}$$

$$1 + \epsilon^2 T_n^2(\omega_{3dB}) = 2$$

$$T_n(\omega_{3dB}) = \frac{1}{\epsilon}$$

For $\omega > 1$,

$$T_n(\omega) = \cosh[n \cosh^{-1}(\omega)]$$

Therefore,

$$\cosh[n \cosh^{-1}(\omega_{3dB})] = \frac{1}{\epsilon}$$

$$\boxed{\omega_{3dB} = \cosh \left[\frac{1}{n} \cosh^{-1} \left(\frac{1}{\epsilon} \right) \right]}$$

9. Butterworth vs Chebyshev: Semester Comparison

Point	Butterworth Filter	Chebyshev Filter
Pass-band response	Maximally flat	Equiripple
Stop-band response	Monotonic	Monotonic for type-I
Transition rate	Slower	Sharper
Ripple	No ripple	Pass-band ripple exists
Pole location	Unit circle	Ellipse
Selectivity	Lower for same order	Higher for same order
Best use	Smooth response	Sharp cut-off requirement

Semester Command

If the question asks “differences between Butterworth and Chebyshev”, write at least six points and draw small response curves. For pole proof questions, do not only write theory; show the circle or ellipse equation.

10. Second-Order Band-Pass Filter Design

10.1. Standard Form

For second-order band-pass filter,

$$H_{BP}(s) = \frac{K Bs}{s^2 + Bs + \omega_0^2}$$

where,

K = pass-band gain

B = bandwidth

ω_0 = center frequency

and,

$$Q = \frac{\omega_0}{B}$$

10.2. Numerical Exam Template

If the question gives

$$\omega_0 = 10 \text{ krad/sec}$$

$$B = 2 \text{ krad/sec}$$

$$K = 10$$

then,

$$\omega_0 = 10^4 \text{ rad/sec}$$

$$B = 2 \times 10^3 \text{ rad/sec}$$

Now,

$$H_{BP}(s) = \frac{K Bs}{s^2 + Bs + \omega_0^2}$$

$$H_{BP}(s) = \frac{10(2000)s}{s^2 + 2000s + (10^4)^2}$$

$$H_{BP}(s) = \frac{20000s}{s^2 + 2000s + 10^8}$$

Quality factor:

$$Q = \frac{\omega_0}{B} = \frac{10^4}{2000}$$

$$Q = 5$$

10.3. Band-Pass Response Diagram

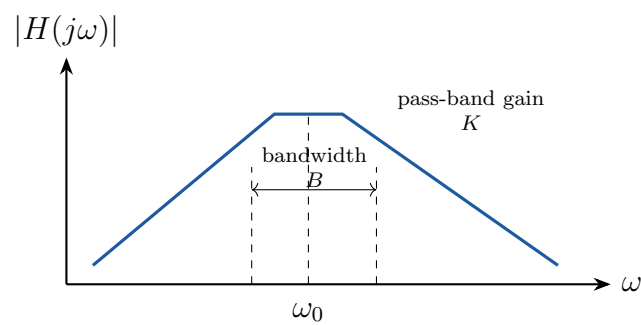


Figure 4: Second-order band-pass filter response.

11. Band-Stop Filter from Band-Pass Filter

A band-stop filter with same center frequency and same pass-band gain can be obtained by subtracting the band-pass output from a direct path.

If,

$$H_{BP}(s) = \frac{KBs}{s^2 + Bs + \omega_0^2}$$

then,

$$H_{BS}(s) = K - H_{BP}(s)$$

$$H_{BS}(s) = K - \frac{KBs}{s^2 + Bs + \omega_0^2}$$

Taking LCM,

$$H_{BS}(s) = \frac{K(s^2 + Bs + \omega_0^2) - KBs}{s^2 + Bs + \omega_0^2}$$

$$H_{BS}(s) = \frac{K(s^2 + \omega_0^2)}{s^2 + Bs + \omega_0^2}$$

For,

$$K = 10, \quad B = 2000, \quad \omega_0 = 10^4$$

$$H_{BS}(s) = \frac{10(s^2 + 10^8)}{s^2 + 2000s + 10^8}$$

11.1. Band-Stop Realization Diagram

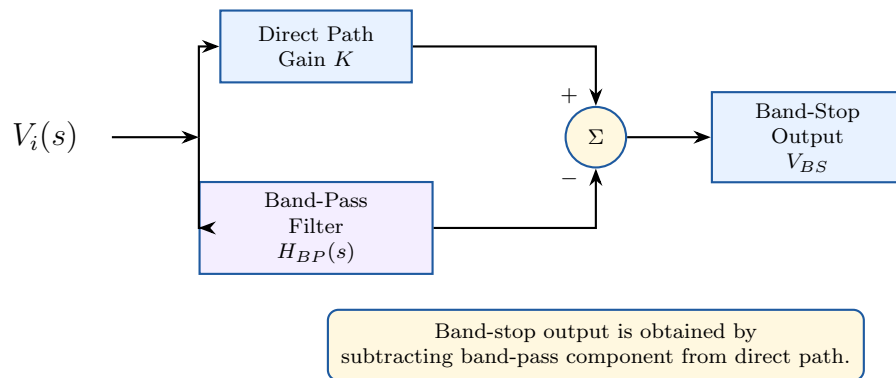


Figure 5: Band-stop filter obtained from a band-pass filter.

12. State Variable Filter

12.1. Definition

A state variable filter is an active filter realized using summers and integrators. It is called a universal filter because low-pass, band-pass, high-pass and band-stop outputs can be obtained from the same circuit.

State variable filter = summer + two integrators + feedback

12.2. Why It is Called Universal Filter

It is called universal because one circuit can provide:

LPF, HPF, BPF and BSF outputs

by taking output from different points.

Must Know

For universal filter questions, always write:

Same circuit gives LP, HP, BP and notch outputs.

This line is compulsory.

12.3. State Variable Block Diagram

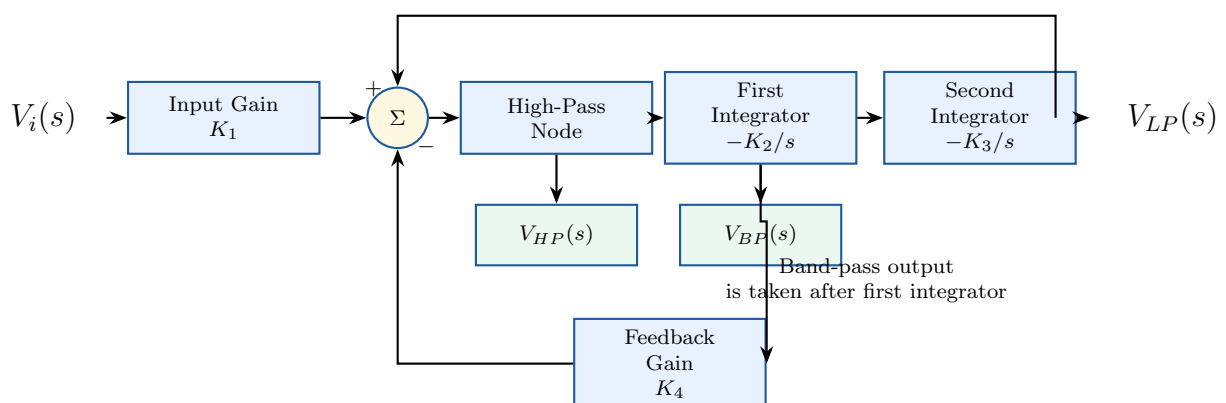


Figure 6: State variable filter showing LP, BP and HP output points.

13. State Variable Transfer Functions

For the state variable filter, the important transfer functions are:

$$H_{HP}(s) = \frac{K_1 s^2}{s^2 + K_2 K_4 s + K_2 K_3}$$

$$H_{BP}(s) = \frac{-K_1 K_2 s}{s^2 + K_2 K_4 s + K_2 K_3}$$

$$H_{LP}(s) = \frac{K_1 K_2 K_3}{s^2 + K_2 K_4 s + K_2 K_3}$$

13.1. Comparison with Standard Denominator

Standard second-order denominator:

$$s^2 + \frac{\omega_0}{Q}s + \omega_0^2$$

State variable denominator:

$$s^2 + K_2 K_4 s + K_2 K_3$$

Therefore,

$$\omega_0^2 = K_2 K_3$$

$$\omega_0 = \sqrt{K_2 K_3}$$

and,

$$\frac{\omega_0}{Q} = K_2 K_4$$

Thus,

$$Q = \frac{\sqrt{K_2 K_3}}{K_2 K_4}$$

For band-pass filter,

$$B = \frac{\omega_0}{Q}$$

so,

$$B = K_2 K_4$$

14. Switched-Capacitor Circuits

14.1. Definition

A switched-capacitor circuit replaces a resistor by a capacitor and switches operated by non-overlapping clock phases.

$$\boxed{\text{Resistor } R \longrightarrow \text{switched capacitor } C_s}$$

The equivalent resistance is

$$\boxed{R_{\text{eq}} = \frac{1}{f_s C_s}}$$

where,

$$f_s = \text{switching frequency}$$

and,

$$C_s = \text{switched capacitor}$$

14.2. Derivation of Equivalent Resistance

During one clock period,

$$T_s = \frac{1}{f_s}$$

the switched capacitor transfers charge

$$\Delta Q = C_s(V_1 - V_2)$$

Average current is charge transferred per unit time:

$$I_{\text{avg}} = \frac{\Delta Q}{T_s}$$

$$I_{\text{avg}} = C_s(V_1 - V_2)f_s$$

For an equivalent resistor,

$$I = \frac{V_1 - V_2}{R_{\text{eq}}}$$

Equating,

$$\frac{V_1 - V_2}{R_{\text{eq}}} = C_s(V_1 - V_2)f_s$$

Cancel $V_1 - V_2$:

$$\frac{1}{R_{\text{eq}}} = C_s f_s$$

Therefore,

$$R_{\text{eq}} = \frac{1}{f_s C_s}$$

14.3. Resistor to Switched-Capacitor Diagram

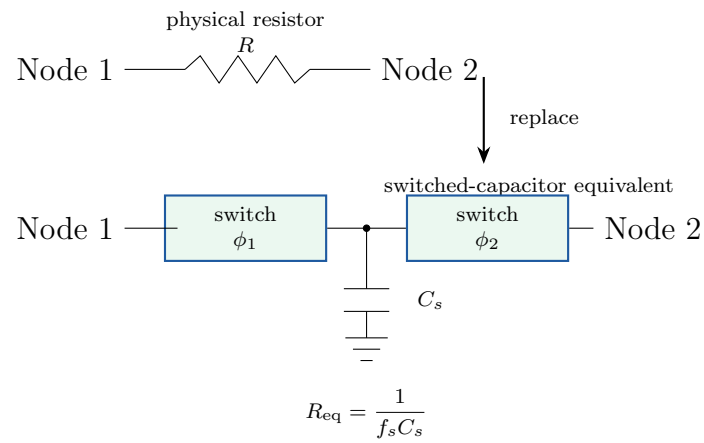


Figure 7: A resistor implemented by a switched capacitor.

15. Advantages of Switched-Capacitor Circuits

1. **Suitable for IC fabrication:** Accurate resistors need large chip area, but capacitors and MOS switches can be fabricated easily.
2. **Large equivalent resistance possible:** Since

$$R_{\text{eq}} = \frac{1}{f_s C_s},$$

large resistance can be realized by using small capacitance and proper clock frequency.

3. **Electronic tunability:** The equivalent resistance depends on clock frequency f_s . So the filter response can be tuned by changing the clock.
4. **Good accuracy:** Capacitor ratios are more accurate in IC technology than resistor values.
5. **Small chip area:** Switched capacitors reduce area compared with large physical resistors.
6. **Useful for active filter realization:** Low-pass, high-pass, band-pass and band-stop active filters can be realized in IC form.

Must Know

For switched-capacitor advantage questions, the strongest line is:

Large resistors are avoided; equivalent resistance is set by clock frequency.

16. Active RC to Switched-Capacitor Conversion

16.1. Basic Rule

Every resistor R_i in the active RC filter is replaced by a switched capacitor C_{Si} :

$$C_{Si} = \frac{1}{R_i f_s}$$

The physical capacitor already present in the original circuit remains unchanged.

Only resistors are replaced. Existing capacitors remain as they are.

16.2. Example

If,

$$R = 10 \text{ k}\Omega$$

and,

$$f_s = 100 \text{ kHz}$$

then,

$$C_s = \frac{1}{R f_s}$$

$$C_s = \frac{1}{(10 \times 10^3)(100 \times 10^3)}$$

$$C_s = \frac{1}{10^9}$$

$$C_s = 1 \text{ nF}$$

16.3. Clean Switched-Capacitor Active Filter Idea

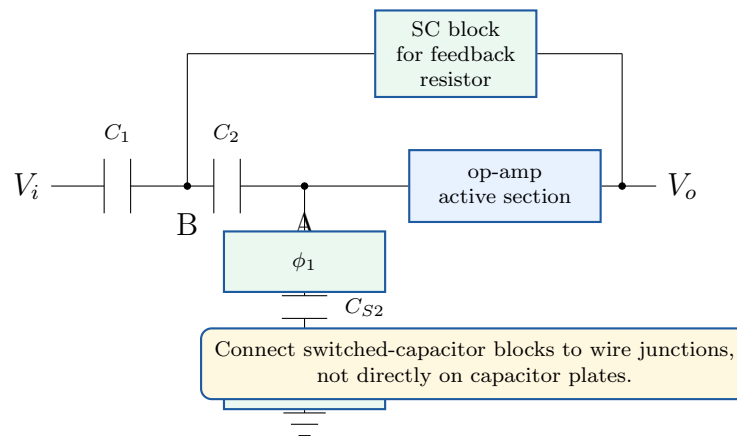


Figure 8: Clean switched-capacitor realization idea for active filter conversion.

17. Low-Pass Filter Design by Butterworth Approximation

17.1. Order Calculation

For Butterworth low-pass filter,

$$A(\omega) = 10 \log_{10} \left[1 + \left(\frac{\omega}{\omega_c} \right)^{2n} \right]$$

If the 3 dB cut-off frequency is given, then ω_c is directly known.

To test order, put $n = 1, 2, 3, \dots$ until both pass-band and stop-band conditions are satisfied.

17.2. Second-Order Butterworth Transfer Function

Normalized second-order Butterworth denominator is

$$s^2 + \sqrt{2}s + 1$$

After frequency scaling,

$$s \rightarrow \frac{s}{\omega_c}$$

So,

$$H(s) = \frac{K\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2}$$

where K is pass-band gain.

17.3. Sallen-Key Low-Pass Conceptual Diagram

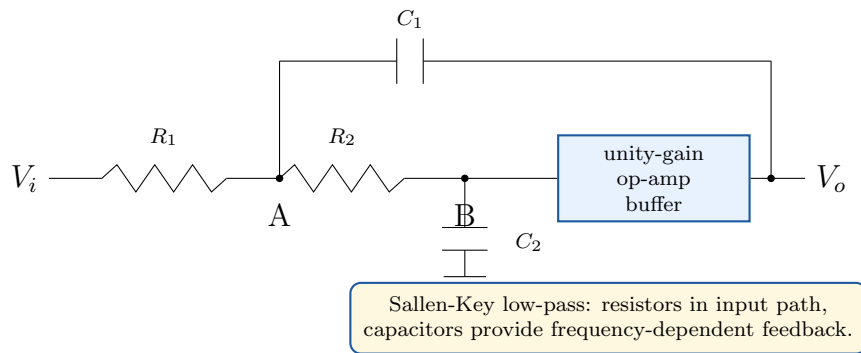


Figure 9: Conceptual Sallen-Key low-pass filter.

18. High-Pass Filter Concept and Switched-Capacitor Version

18.1. Second-Order High-Pass Standard Form

$$H_{HP}(s) = \frac{K s^2}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2}$$

At low frequency $s \rightarrow 0$,

$$H_{HP}(0) = 0$$

So DC is blocked.

At high frequency $s \rightarrow \infty$,

$$H_{HP}(s) \rightarrow K$$

So high frequencies are passed.

18.2. Active RC High-Pass Conceptual Diagram

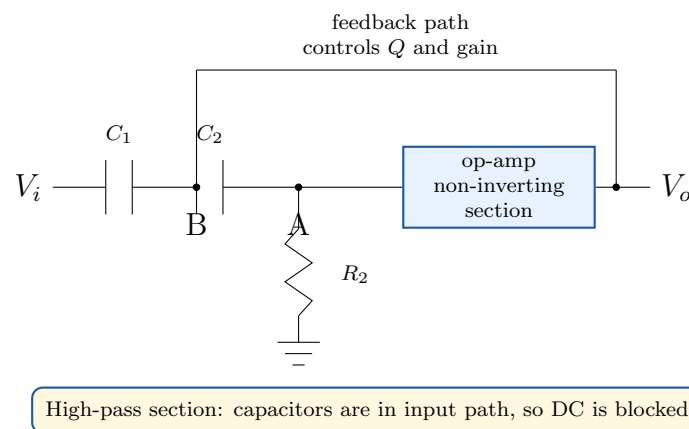


Figure 10: Conceptual second-order active high-pass filter before switched-capacitor conversion.

19. Exam Design Routine

19.1. Routine for Band-Pass Numerical

Semester Command

For band-pass design:

$$\boxed{\text{Read } \omega_0, B, K} \rightarrow \boxed{Q = \omega_0/B} \rightarrow \boxed{H_{BP}(s) = \frac{KBs}{s^2 + Bs + \omega_0^2}}$$

19.2. Routine for Band-Stop Numerical

Semester Command

For band-stop from band-pass:

$$\boxed{H_{BS}(s) = K - H_{BP}(s)}$$

or directly,

$$\boxed{H_{BS}(s) = \frac{K(s^2 + \omega_0^2)}{s^2 + Bs + \omega_0^2}}$$

19.3. Routine for Chebyshev Order

For Chebyshev order,

$$\boxed{n \geq \frac{\cosh^{-1} \left[\sqrt{\frac{10^{A_s/10} - 1}{10^{A_p/10} - 1}} \right]}{\cosh^{-1} \left(\frac{\omega_s}{\omega_p} \right)}}$$

Ripple factor:

$$\boxed{\epsilon = \sqrt{10^{A_p/10} - 1}}$$

19.4. Routine for Switched-Capacitor Conversion

$$\boxed{C_{Si} = \frac{1}{R_i f_s}}$$

If all resistors have same value R , then all switched capacitors are same:

$$C_{S1} = C_{S2} = C_{S3} = \cdots = \frac{1}{Rf_s}$$

20. One-Page Formula Bank

$$|H_B(j\omega)|^2 = \frac{1}{1 + \omega^{2n}}$$

$$|H_C(j\omega)|^2 = \frac{1}{1 + \epsilon^2 T_n^2(\omega)}$$

$$\epsilon = \sqrt{10^{A_p/10} - 1}$$

$$H_{LP}(s) = \frac{K\omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$H_{HP}(s) = \frac{Ks^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$H_{BP}(s) = \frac{KBs}{s^2 + Bs + \omega_0^2}$$

$$H_{BS}(s) = \frac{K(s^2 + \omega_0^2)}{s^2 + Bs + \omega_0^2}$$

$$Q = \frac{\omega_0}{B}$$

$$R_{eq} = \frac{1}{f_s C_s}$$

$$C_s = \frac{1}{R f_s}$$

21. How to Recognise Filter Questions in Semester

Question Language	What You Must Write
“Butterworth poles unit circle”	Start from $s_k = -\sin \theta + j \cos \theta$, show $ s_k = 1$, draw circle.
“Chebyshev poles ellipse”	Use $s_k = -\sinh \alpha \sin \theta + j \cosh \alpha \cos \theta$, derive ellipse equation.
“Why Chebyshev $-3\text{ dB} > 1$ ”	Show $ H(j1) ^2 = 1/(1+\epsilon^2) > 1/2$, hence $\omega_{3dB} > 1$.
“State variable universal filter”	Draw summer + two integrators; mention LP, BP, HP and notch outputs.
“Band-pass transfer function”	Use $H_{BP}(s) = KBs/(s^2 + Bs + \omega_0^2)$.
“Band-stop from band-pass”	Use $H_{BS}(s) = K - H_{BP}(s)$.
“Switched capacitor advantages”	Write $R_{eq} = 1/(f_s C_s)$, IC fabrication, clock tunability.
“Equivalent switched capacitor”	Calculate $C_s = 1/(Rf_s)$, replace each resistor by SC block.

22. Final Rapid Revision for Phase 3

22.1. If Only 30 Minutes Are Left

Revise in this order:

1. Butterworth pole proof.
2. Chebyshev ellipse proof.
3. Standard LP, HP, BP, BS transfer functions.
4. $Q = \omega_0/B$ and band-pass numerical.
5. State variable universal filter diagram.
6. Switched-capacitor formula $R_{eq} = 1/(f_s C_s)$.
7. Advantages of switched-capacitor filters.

22.2. Most Probable Diagram Commands

1. Butterworth unit circle.
2. Chebyshev ellipse.
3. State variable filter block diagram.
4. Band-stop from band-pass summing diagram.
5. Switched-capacitor resistor replacement.
6. Active RC to switched-capacitor conversion.

Do Not Write This Mistake

Do not draw crowded diagrams. If the diagram is large, draw it in two stages. Never place labels on wires. Always connect feedback to a wire junction, not directly on capacitor plates or resistor bodies.

Phase 3 Completed: Active Filter and Switched-Capacitor Mastery